



## Indian Institute of Technology Bhubaneswar

School of Infrastructure

Session: Spring 2026

Finite Element Method (CE6L314)

Solution of Class Test 1

1. Consider the boundary value problem

$$-u''(x) = f(x), \quad x \in (0, 1)$$

$$u(0) = 0, \quad u(1) = 0$$

- (a) Show that the problem is equivalent to minimizing a functional.
- (b) Derive the energy functional  $J(u)$ .
- (c) Prove that the Euler–Lagrange equation of  $J(u)$  gives the original differential equation.
- (d) Specify the admissible function space.

**Solution:**

**(a) Equivalent minimization problem**

Multiply by a test function  $v(x)$  with  $v(0) = v(1) = 0$ :

$$\int_0^1 (-u'')v \, dx = \int_0^1 f v \, dx.$$

Integrating by parts:

$$\int_0^1 u' v' \, dx - [u' v]_0^1 = \int_0^1 f v \, dx.$$

Since  $v(0) = v(1) = 0$ , boundary terms vanish:

$$\int_0^1 u' v' \, dx = \int_0^1 f v \, dx.$$

Define

$$a(u, v) = \int_0^1 u' v' \, dx, \quad \ell(v) = \int_0^1 f v \, dx.$$

The problem is equivalent to minimizing

$$J(u) = \frac{1}{2} a(u, u) - \ell(u).$$

**(b) Energy functional**

$$J(u) = \frac{1}{2} \int_0^1 |u'|^2 \, dx - \int_0^1 f u \, dx$$

**(c) Euler–Lagrange equation**

Let  $u_\varepsilon = u + \varepsilon v$ .

$$\frac{d}{d\varepsilon} J(u + \varepsilon v) = \int_0^1 u' v' dx - \int_0^1 f v dx.$$

Setting equal to zero:

$$\int_0^1 u' v' dx = \int_0^1 f v dx.$$

Integrating by parts again gives:

$$-u'' = f.$$

**(d) Admissible function space**

$$\boxed{V = H_0^1(0, 1)}$$

2. Consider

$$\begin{aligned} -u'' &= 1, \quad x \in (0, 1) \\ u(0) &= u(1) = 0 \end{aligned}$$

Assume the trial function

$$u_N(x) = a_1 x(1 - x)$$

(a) Write the energy functional.

(b) Substitute the trial function into the functional.

(c) Minimize with respect to  $a_1$ .

(d) Compare with the exact solution.

**Solution:**

(a) Energy functional:

$$J(u) = \frac{1}{2} \int_0^1 (u')^2 dx - \int_0^1 u dx.$$

(b) Derivative:

$$u'_N = a_1(1 - 2x).$$

$$\int_0^1 (u'_N)^2 dx = a_1^2 \int_0^1 (1 - 2x)^2 dx = a_1^2 \frac{1}{3}.$$

$$\int_0^1 u_N dx = a_1 \int_0^1 x(1 - x) dx = \frac{a_1}{6}.$$

Thus

$$J(a_1) = \frac{a_1^2}{6} - \frac{a_1}{6}.$$

(c) Minimizing:

$$\frac{dJ}{da_1} = \frac{a_1}{3} - \frac{1}{6} = 0 \Rightarrow a_1 = \frac{1}{2}.$$

(d) Approximate solution:

$$u_N = \frac{1}{2} x(1 - x).$$

Exact solution:

$$u(x) = \frac{1}{2}x(1-x).$$

So, for this case, an approximate solution exactly matches the exact one.

3. Consider

$$-u''(x) = f(x), \quad x \in (0, 1)$$

$$u(0) = 0, \quad u'(1) = g$$

(a) Derive the weak form.

(b) Show how the Neumann boundary condition appears naturally.

(c) Explain why the boundary condition  $u'(1) = g$  is called a natural boundary condition.

**Solution:**

(a) Multiply by  $v(x)$  with  $v(0) = 0$ :

$$\int_0^1 (-u'')v dx = \int_0^1 f v dx.$$

Integration by parts:

$$\int_0^1 u' v' dx - [u' v]_0^1 = \int_0^1 f v dx.$$

(b) Since  $v(0) = 0$ ,

$$\int_0^1 u' v' dx = \int_0^1 f v dx + g v(1).$$

The Neumann condition appears naturally from the boundary term.

(c) It is called a **natural boundary condition** because it arises automatically from the weak formulation.

4. For an element,  $\Omega^{(e)} = [x_i, x_{i+1}]$ :

(a) Derive quadratic shape functions  $N_1^{(e)}$ ,  $N_2^{(e)}$  and  $N_3^{(e)}$ .

(b) Show the partition of unity property.

(c) Why do the shape functions satisfy the partition of unity property?

**Solution:**

Consider the finite element

$$\Omega^{(e)} = [x_i, x_{i+1}]$$

with three nodes:

$$x_i, \quad x_{i+\frac{1}{2}} = \frac{x_i + x_{i+1}}{2}, \quad x_{i+1}.$$

(a) Derivation of Quadratic Shape Functions:

Quadratic shape functions satisfy the interpolation conditions:

$$N_1^{(e)}(x_i) = 1, \quad N_1^{(e)}(x_{i+\frac{1}{2}}) = 0, \quad N_1^{(e)}(x_{i+1}) = 0,$$

$$N_2^{(e)}(x_i) = 0, \quad N_2^{(e)}(x_{i+\frac{1}{2}}) = 1, \quad N_2^{(e)}(x_{i+1}) = 0,$$

$$N_3^{(e)}(x_i) = 0, \quad N_3^{(e)}(x_{i+\frac{1}{2}}) = 0, \quad N_3^{(e)}(x_{i+1}) = 1$$

Using the above conditions, quadratic shape functions can be derived in terms of polynomials as follows:

$$N_1^{(e)}(x) = \frac{(x - x_{i+\frac{1}{2}})(x - x_{i+1})}{(x_i - x_{i+\frac{1}{2}})(x_i - x_{i+1})},$$

$$N_2^{(e)}(x) = \frac{(x - x_i)(x - x_{i+1})}{(x_{i+\frac{1}{2}} - x_i)(x_{i+\frac{1}{2}} - x_{i+1})},$$

$$N_3^{(e)}(x) = \frac{(x - x_i)(x - x_{i+\frac{1}{2}})}{(x_{i+1} - x_i)(x_{i+1} - x_{i+\frac{1}{2}})}.$$

Reference Element Form:

Let the reference coordinate be  $\xi \in [-1, 1]$ .

Nodes:

$$\xi_1 = -1, \quad \xi_2 = 0, \quad \xi_3 = 1.$$

Shape functions become

$$N_1(\xi) = \frac{\xi(\xi - 1)}{2},$$

$$N_2(\xi) = 1 - \xi^2,$$

$$N_3(\xi) = \frac{\xi(\xi + 1)}{2}.$$

**Plot the shape functions.**

(b) Partition of Unity:

Sum the three shape functions:

$$N_1(\xi) + N_2(\xi) + N_3(\xi) = \frac{\xi(\xi - 1)}{2} + (1 - \xi^2) + \frac{\xi(\xi + 1)}{2}.$$

Simplifying:

$$= \frac{\xi^2 - \xi + \xi^2 + \xi}{2} + 1 - \xi^2 = \frac{2\xi^2}{2} + 1 - \xi^2 = \xi^2 + 1 - \xi^2 = 1.$$

$$\boxed{N_1 + N_2 + N_3 = 1}$$

Hence, quadratic shape functions satisfy the partition of unity property.

**Show the partition of unity at any point in the plot of the combination of all the shape functions.**

(c) Why Partition of Unity Holds:

The partition of unity property holds because

- The shape functions are constructed in such a way that the rigid body motion of a body can be modeled.
- The Lagrange polynomials exactly reproduce constant functions.
- The constant function  $u(x) = 1$  is interpolated exactly.
- Therefore, the sum of all shape functions must equal 1.

Mathematically, since interpolation satisfies

$$1 = \sum_{a=1}^3 1 \cdot N_a(x),$$

we obtain

$$\sum_{a=1}^3 N_a(x) = 1.$$

5. Consider

$$-u'' = 1, \quad x \in (0, 1)$$

$$u(0) = 0, \quad u(1) = 0$$

Using a uniform mesh with two and three equal elements:

- (a) Construct the global hat basis functions.
- (b) Write the finite-dimensional approximation space.
- (c) Derive the stiffness matrix.
- (d) Derive the load vector.
- (e) Solve the resulting algebraic system.
- (f) Compare with the exact solution.

**Solution:**

The exact solution is obtained by integration:

$$u'' = -1 \Rightarrow u(x) = \frac{1}{2}x(1-x).$$

Case 1: Two Equal Elements

$$h = \frac{1}{2}, \quad x_0 = 0, \quad x_1 = \frac{1}{2}, \quad x_2 = 1.$$

There is one interior node.

(a) Global Hat Function

$$\phi_1(x) = \begin{cases} 2x, & 0 \leq x \leq \frac{1}{2}, \\ 2(1-x), & \frac{1}{2} \leq x \leq 1. \end{cases}$$

(b) Approximation Space

$$V_h = \text{span}\{\phi_1\}.$$

(c) Stiffness Matrix

$$K_{11} = \int_0^1 (\phi_1')^2 dx.$$

$$\phi_1' = \begin{cases} 2, & 0 < x < \frac{1}{2}, \\ -2, & \frac{1}{2} < x < 1. \end{cases}$$

$$K_{11} = 4 \left( \frac{1}{2} \right) + 4 \left( \frac{1}{2} \right) = 4.$$

(d) Load Vector

$$F_1 = \int_0^1 \phi_1 dx.$$

Area of triangle:

$$F_1 = \frac{1}{2}.$$

(e) Algebraic System

$$4U_1 = \frac{1}{2} \Rightarrow U_1 = \frac{1}{8}.$$

(f) Comparison

Exact at  $x = \frac{1}{2}$ :

$$u\left(\frac{1}{2}\right) = \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{8}.$$

Exact nodal agreement.

Case 2: Three Equal Elements

$$h = \frac{1}{3}, \quad x_0 = 0, \quad x_1 = \frac{1}{3}, \quad x_2 = \frac{2}{3}, \quad x_3 = 1.$$

Two interior nodes.

(a) Global Hat Functions

$$\phi_1(x) = \begin{cases} 3x, & 0 \leq x \leq \frac{1}{3}, \\ 3\left(\frac{2}{3} - x\right), & \frac{1}{3} \leq x \leq \frac{2}{3}, \\ 0, & \text{otherwise.} \end{cases}$$

$$\phi_2(x) = \begin{cases} 3\left(x - \frac{1}{3}\right), & \frac{1}{3} \leq x \leq \frac{2}{3}, \\ 3(1 - x), & \frac{2}{3} \leq x \leq 1, \\ 0, & \text{otherwise.} \end{cases}$$

**Plot the hat functions.**

(b) Approximation Space

$$V_h = \text{span}\{\phi_1, \phi_2\}.$$

(c) Stiffness Matrix

For uniform mesh:

$$K = \frac{1}{h} \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}.$$

Since  $h = \frac{1}{3}$ :

$$K = 3 \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} = \begin{bmatrix} 6 & -3 \\ -3 & 6 \end{bmatrix}.$$

(d) Load Vector

For constant load  $f = 1$ :

$$F_i = \int_0^1 \phi_i dx.$$

Each hat function has area:

$$F_1 = F_2 = \frac{1}{3}.$$

Thus

$$F = \begin{bmatrix} \frac{1}{3} \\ \frac{1}{3} \end{bmatrix}.$$

(e) Algebraic System

$$\begin{bmatrix} 6 & -3 \\ -3 & 6 \end{bmatrix} \begin{bmatrix} U_1 \\ U_2 \end{bmatrix} = \begin{bmatrix} \frac{1}{3} \\ \frac{1}{3} \end{bmatrix}.$$

Solving:

$$U_1 = U_2 = \frac{1}{9}.$$

(f) Comparison with Exact Solution

Exact solution:

$$u(x) = \frac{1}{2}x(1-x).$$

Exact values:

$$u\left(\frac{1}{3}\right) = \frac{1}{2} \cdot \frac{1}{3} \cdot \frac{2}{3} = \frac{1}{9},$$

$$u\left(\frac{2}{3}\right) = \frac{1}{9}.$$

Again, exact nodal agreement.

**Plot the numerical results with two and three elements and compare them with the exact solution for all  $x \in (0, 1)$ .**

6. Consider

$$-u'' = 1, \quad x \in (0, 1)$$

$$u(0) = 0, \quad u(1) = 0$$

Using non-uniform mesh:

$$x_0 = 0, \quad x_1 = 0.3, \quad x_2 = 1$$

(a) Construct the global hat functions.

(b) Starting from the weak form, derive the algebraic system of equations in matrix-vector form.

(c) Derive stiffness matrix entries explicitly.

**Solution:**

Let

$$h_1 = x_1 - x_0 = 0.3, \quad h_2 = x_2 - x_1 = 0.7.$$

Since the boundary values are zero, there is only one unknown degree of freedom at node  $x_1$ .

(a) The global basis (hat) function associated with node  $x_1$  is

$$\phi_1(x) = \begin{cases} \frac{x}{0.3}, & 0 \leq x \leq 0.3, \\ \frac{1-x}{0.7}, & 0.3 \leq x \leq 1. \end{cases}$$

**Plot the hat functions.**

The finite-dimensional space is

$$V_h = \text{span}\{\phi_1\}.$$

(b) The weak form of the problem is:

$$\int_0^1 u' v' dx = \int_0^1 v dx \quad \forall v \in H_0^1(0,1).$$

Approximate the solution as

$$u_h = U_1 \phi_1.$$

Choose  $v = \phi_1$ :

$$\int_0^1 u_h' \phi_1' dx = \int_0^1 \phi_1 dx.$$

Substitute:

$$U_1 \int_0^1 (\phi_1')^2 dx = \int_0^1 \phi_1 dx.$$

This yields the matrix system:

$$\boxed{\begin{bmatrix} K_{11} \end{bmatrix} \begin{bmatrix} U_1 \end{bmatrix} = \begin{bmatrix} F_1 \end{bmatrix}}$$

where

$$K_{11} = \int_0^1 (\phi_1')^2 dx, \quad F_1 = \int_0^1 \phi_1 dx.$$

(c) Derivative of  $\phi_1$ :

$$\phi_1' = \begin{cases} \frac{1}{0.3}, & 0 < x < 0.3, \\ -\frac{1}{0.7}, & 0.3 < x < 1. \end{cases}$$

Thus

$$\begin{aligned} K_{11} &= \int_0^{0.3} \left( \frac{1}{0.3} \right)^2 dx + \int_{0.3}^1 \left( \frac{-1}{0.7} \right)^2 dx. \\ &= \frac{1}{0.09}(0.3) + \frac{1}{0.49}(0.7). \\ &= \frac{0.3}{0.09} + \frac{0.7}{0.49} = 3.3333 + 1.4286 = 4.7619. \end{aligned}$$

Load Vector Entry

$$F_1 = \int_0^{0.3} \frac{x}{0.3} dx + \int_{0.3}^1 \frac{1-x}{0.7} dx.$$

First integral:

$$\frac{1}{0.3} \cdot \frac{x^2}{2} \Big|_0^{0.3} = \frac{0.09}{0.6} = 0.15.$$

Second integral:

$$\frac{1}{0.7} \left[ x - \frac{x^2}{2} \right]_{0.3}^1 = \frac{1}{0.7} (0.245) = 0.35.$$

Thus

$$F_1 = 0.15 + 0.35 = 0.5.$$

Final Algebraic Equation

$$4.7619 U_1 = 0.5$$

$$U_1 = 0.105.$$

Final Approximate Solution

$$u_h(x) = 0.105 \phi_1(x).$$